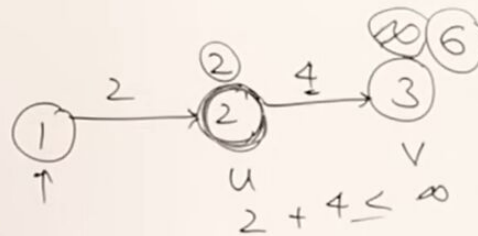
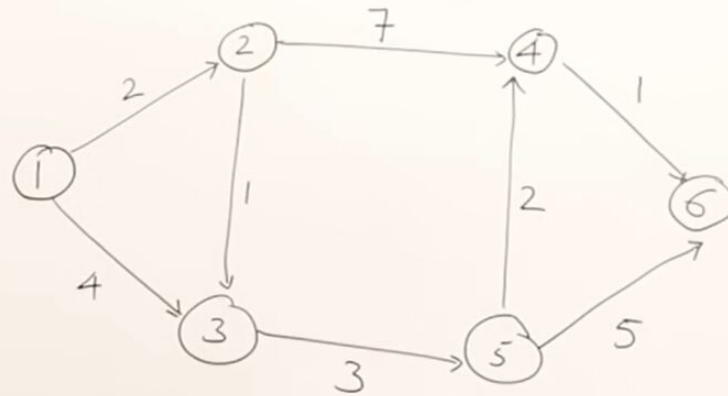


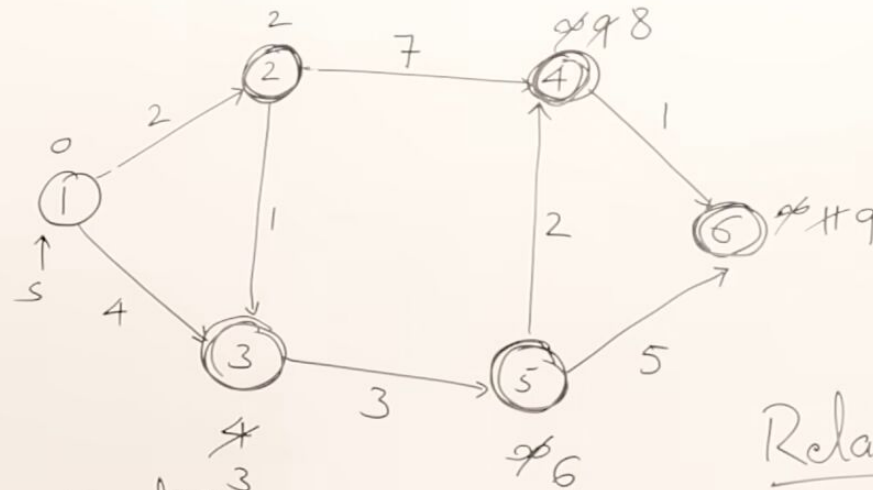
Dijkstra Algorithm



Relaxation
if $(d[u] + c(u,v) < d[v])$
 $d[v] = d[u] + c(u,v)$



Dijkstra Algorithm



v	d[v]
2	2
3	3
4	8
5	6
6	9

$$n = |V| \quad |V|$$

$$n \times n = n^2 \quad O(|V|^2)$$

Relaxation

$$\text{if } (d[u] + c(u,v) < d[v])$$

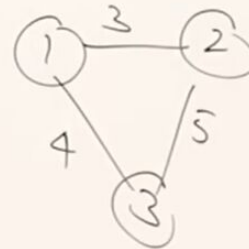
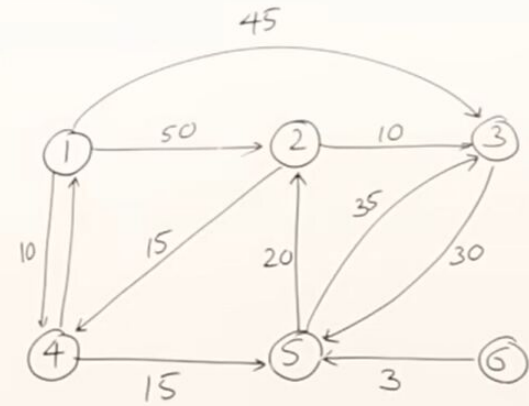
$$d[v] = d[u] + c(u,v)$$



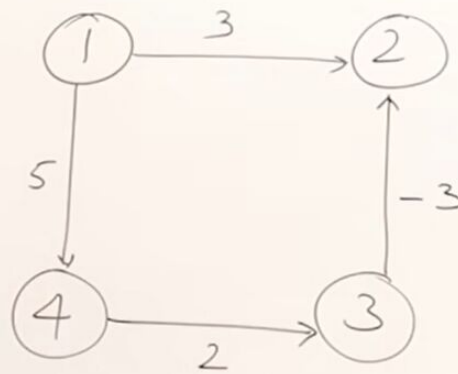
Dijkstra Algorithm

starting vertex 1

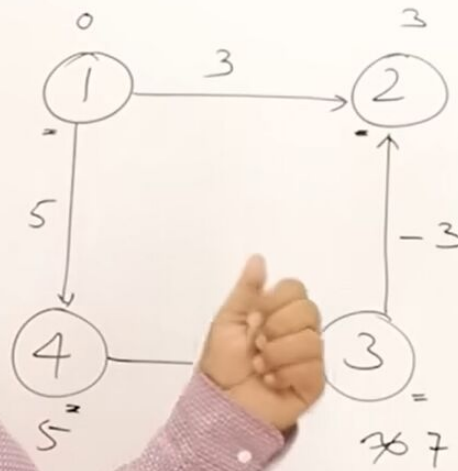
selected vertex	2	3	4	5	6
4	50	45	(10)	∞	∞
5	50	45	(10)	(25)	∞
2	(45)	45	(10)	(25)	∞
3	(45)	(45)	(10)	(25)	∞
6	(45)	(45)	(10)	(25)	(20)



Dijkstra Algorithm



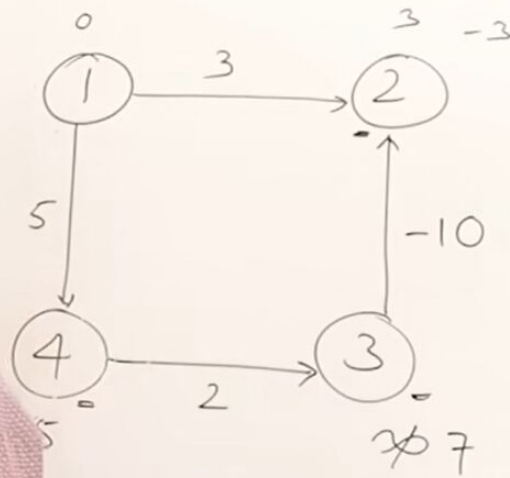
Dijkstra Algorithm



worked even when there is a negative weight



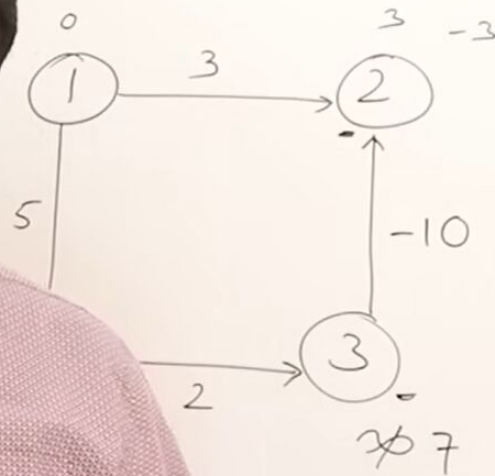
Dijkstra Algorithm



answer is minus 3 but distra has given us the wrong answer



Dijkstra Algorithm



have a Negative Edge so Dijkstra algorithm may work or may not work

